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Prof. Fabrice Theoleyre, Prof. Oana Iova, and Prof. Fabrice Valois
ICUBE & CITI Laboratories

Subject: Application for PhD Position – Digital Twins for Wireless Networks (PC4)

Dear Prof. Theoleyre, Prof. Iova, and Prof. Valois,

I am applying for the PhD position on Digital Twins for Wireless Networks with great enthusiasm. As an engineer specializing in Edge-Optimized AI and distributed architectures, the challenge of conceptualizing frugal data collection mechanisms for time-variant wireless environments aligns perfectly with my expertise in building intelligent, resource-constrained systems.

My background directly addresses your core objective: accurately mirroring low-power wireless networks without crippling network overhead. While developing an Industrial Gas Turbine Digital Twin, I recognized the severe limitations of exhaustive telemetry. To resolve this, I architected Temporal Convolutional Networks (Edge-TCN) and lightweight Mamba-SSM sequence models directly on edge hardware. This allowed for highly accurate predictive state inference using minimal, intelligently sampled data streams. Applying these exact methodologies to wireless network telemetry can significantly reduce the data volume required for synchronization. Furthermore, my hands-on experience using Reinforcement Learning and Graph Neural Networks (GNNs) provides a strong foundation for exploring “what-if” scenarios and automating closed-loop control decisions, such as routing adjustments before physical failures occur.

I am eager to combine my programming proficiency in Python and C with my experience in autonomous systems to contribute effectively to the Artemis framework and the SLICES-FR platform. Please find attached my resume and transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui